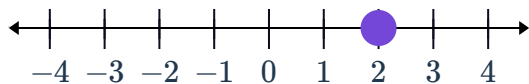


Number lines

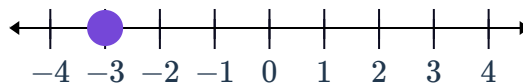


1 State the integer where the point is plotted on the following number lines:

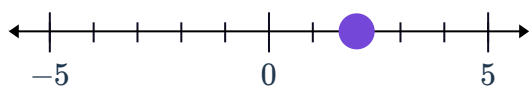
a



b



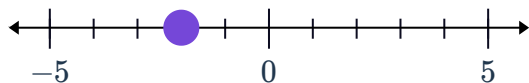
c



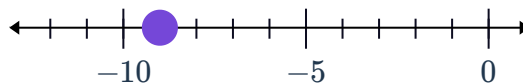
d



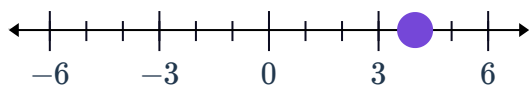
e



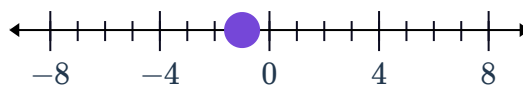
f



g



h



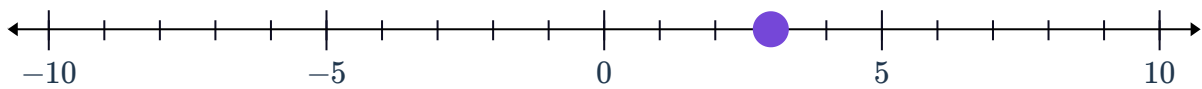
i



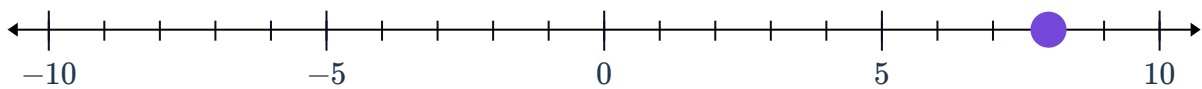
j



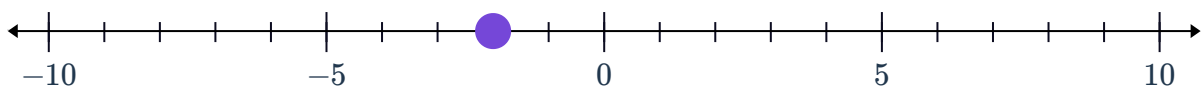
- 2 State the number that is 6 units to the right of the plotted point:



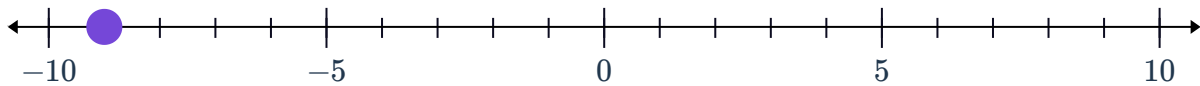
- 3 State the number that is 7 units to the left of the plotted point:



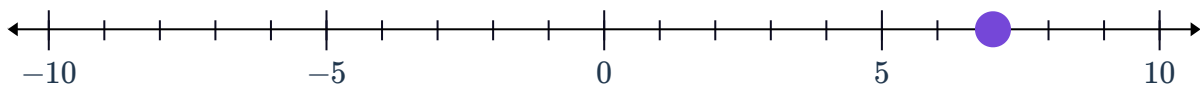
- 4 State the number that is 6 units to the left of the plotted point:



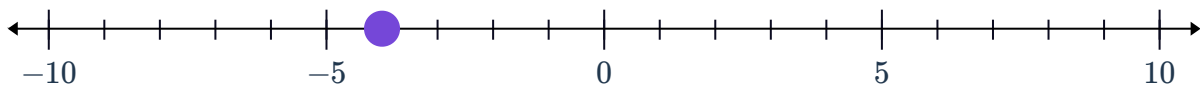
- 5 State the number that is 7 units to the right of the plotted point:



- 6 State the number that is 9 units to the left of the plotted point:



- 7 State the number that is 7 units to the right of the plotted point:



Integers

- 8 Describe how to tell if two integers on a number line are opposites.

9 For each of the following number lines:

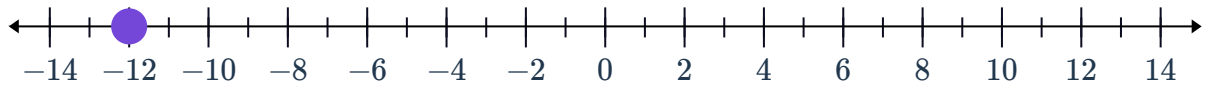


- i State the number that has been plotted.
- ii State the opposite of the number that has been plotted.

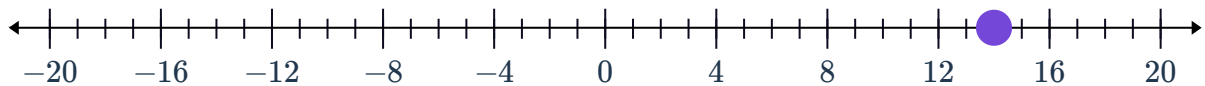
a



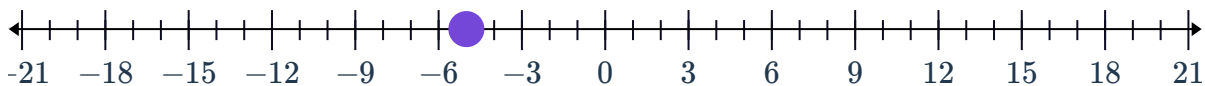
b



c



d



10 Consider the integer 8.



- a State its opposite.
- b True or False: "The opposite of a number is always greater than the number itself."

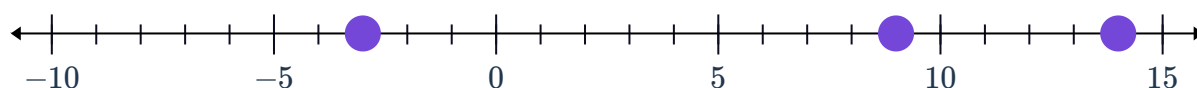
11 State the number that is the opposite of 0.

12 The opposite of a number is -16 . State the original number.

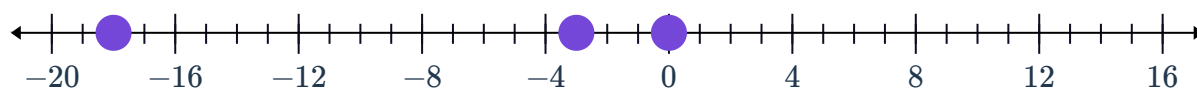
Comparison and order of integers

13 State the greatest number plotted on the following number lines:

a



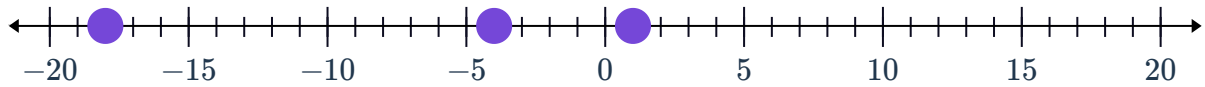
b



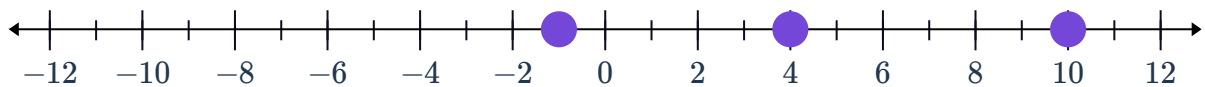
14 State the smallest number plotted on the following number lines:



a



b



15 State the greatest number in each set:

a 13, -14, 2

b -7, -17, -12

c 8, -8, 7

d 35, -36, -40

16 State the smallest number in each set:

a -14, -4, -21

b -7, -24, 10

c 1, -1, 5

d 34, -36, -2

17 Arrange the following numbers in ascending order: 10, -25, 16, -11, 27

18 Arrange the following numbers in descending order: 3, -13, 11, -9, 15

19 For each pair of temperatures, state the higher temperature:

a -19°C or 4°C

b -3°C or -17°C

20 For each pair of temperatures, state the lower temperature:

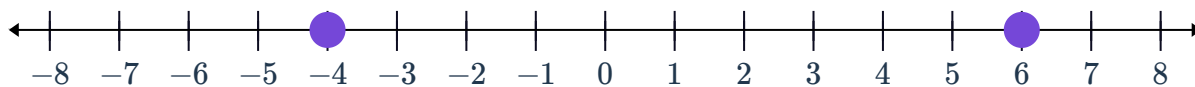
a -15°C or -3°C

b -10°C or 1°C

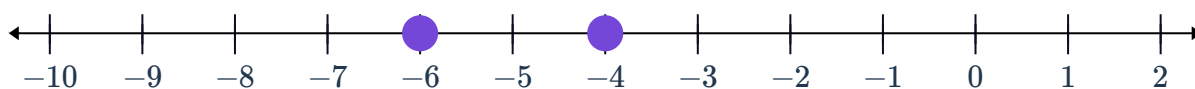
21 Write an inequality that compares the two numbers plotted on the following number lines:



a



b



22 For each of the following inequalities:

i Write the inequality in words.

ii State if the inequality is true or false.

a $-7 < -2$

b $7 > -5$

c $0 > -1$

d $23 < -47$

23 Identify the inequality sign, $<$ or $>$, that should appear between each pair of numbers to make each statement true:

a $7 \square -4$

b $-9 \square 9$

c $0 \square -8$

d $-2 \square -5$

e $39 \square -72$

f $-12 \square -14$

g $-45 \square 22$

h $-1 \square -30$

Applications

24 Consider the following statement: "Arriving 14 minutes late."

a State the opposite of "Arriving 14 minutes late".

b Suppose "Arriving 14 minutes late" is represented by the number 14. State the integer that should represent its opposite.

25 Consider the following statement: "Cycling 8 km due south."



- a** State the opposite of "Cycling 8 km due south".
- b** Suppose "Cycling 8 km due south" is represented by the number 8. State the integer that should represent its opposite.

26 A gain of \$400 can be represented by the integer 400.

- a** State whether the following would represent the opposite of this situation:

- i** Gaining another \$400.
- ii** Losing \$400.
- iii** Losing \$800.

- b** State the integer that represents your answer in part (a).

27 State the integer that represents the opposite of the following:

- a** The statement "Descending 2 floors" is represented by the number -2 .
- b** The statement "A weight gain of 11 kg" is represented by the number 11.
- c** A temperature of 10 degrees below freezing is represented by the integer -10 .

28 A temperature drop of 14°C can be represented by the integer -14 .

- a** State an integer that represents the opposite of a temperature drop of 14°C .
- b** State an integer that represents the opposite of the opposite of a temperature drop of 14°C .

29 A mountain peak is measured to have a height of 5100 m above sea level. An ocean trench is found that is the same distance below sea level.

- a** If the height above sea level of the mountain peak is represented by the number 5100, state the integer that represents the location of the trench.
- b** In this scenario, state what the integer 0 represents.

30 For each of the following scenarios, write  an inequality that compares the two quantities:

- a** Yuri has \$8 in savings. Kenneth has \$13 saved.
- b** Christa's account balance is $-\$10$. Ray's account balance is $-\$17$.
- c** Elizabeth and Roxanne both buy some tickets in the raffle at the local fete. After counting up their winnings, Elizabeth has made \$86, but Roxanne has lost \$146.
- d** Paul has read 27 books this year. His sister Eileen has read 17 books.
- e** Quiana has made \$78.79 this week in her part time job, and has spent \$35.33 on food in that time.
- f** The melting point of krypton is -157°C . The melting point of radon is -71°C .